

APPLIED OPERATING SYSTEM



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MODULE 6

PROTECTION AND SECURITY

OBJECTIVES

Upon completion of this module, the student would be able to:

- Identify the role of the operating system with regard to system security;
- Define the key properties to system survivability;
- Discuss the effects of system security practices on overall system performance;
- Give the levels of system security that can be implemented and the threats posed by evolving technologies;
- Enumerate the different types of security breaches;
- Identify system vulnerabilities and how to protect our system from them;



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ROLE OF OPERATING SYSTEM IN SECURITY



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ROLE OF THE OPERATING SYSTEM IN SECURITY

Operating system plays a key role in computer system security.

- Any vulnerability at the operating system level opens the entire system to attack
- The more complex and powerful the operating system, the more likely it is to have vulnerabilities to attack

System administrators must be on guard to arm their operating systems with all available defenses against attack

SYSTEM SURVIVABILITY

System Survivability is the capability of a system to fulfill its mission, in a timely manner, in the presence of attacks, failures, or accidents

Key properties of survivable systems:

- Resistance to attacks
- Recognition of attacks and resulting damage
- Recovery of essential services after an attack
- Adaptation and evolution of system defense mechanisms to lessen future attacks



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SYSTEM SURVIVABILITY

Key Property	Description	Example Strategies
Resistance to attacks	Strategies for repelling attacks	Authentication Access controls Encryption Message filtering System diversification Functional isolation
Recognition of attacks and damage	Strategies for detecting attacks and evaluating damage	Intrusion detection Integrity checking
Recovery of essential and full services after attack	Strategies for limiting damage, restoring compromised information or functionality, maintaining or restoring essential services within mission time constraints, restoring full services	Redundant components Data replication System backup and restoration Contingency planning
Adaptation and evolution to reduce effectiveness of future attacks	Strategies for improving system survivability based on knowledge gained from intrusions	Intrusion recognition patterns

Four key properties of a survivable system



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LEVELS OF PROTECTION

System administrator must evaluate the risk of intrusion for each computer configuration, which in turn depends on the level of connectivity given to the system

Configuration	Ease of Protection	Relative Risk	Vulnerabilities
Single computer (without e-mail or Internet)	High	Low	Compromised passwords, viruses
LAN connected (without Internet)	Medium	Medium	Sniffers, spoofing (+passwords, viruses)
LAN connected (with Internet)	Low	High	E-mail, Web servers, FTP, Telnet (+sniffers, spoofing, passwords, viruses)

A simplified comparison of security protection required for three typical computer configurations



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BACKUP AND RECOVERY

- Backup and recovery policies are essential for most computing systems
- Many system managers use a layered backup schedule
- Backups, with one set stored off-site, are crucial to disaster recovery
- Written policies and procedures and regular user training are essential elements of system management



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BACKUP AND RECOVERY

- **Written security procedures** should recommend:
 - Frequent password changes
 - Reliable backup procedures
 - Guidelines for loading new software
 - Compliance with software licenses
 - Network safeguards
 - Guidelines for monitoring network activity
 - Rules for terminal access



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SYSTEM SECURITY BREACHES AND SYSTEM PROTECTION



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SECURITY BREACHES

- A gap in system security can be malicious or not
- Intrusions can be classified as:
 - Due to uneducated users and unauthorized access to system resources
 - Purposeful disruption of the system's operation
 - Purely accidental

Examples: Hardware malfunctions, undetected errors in OS or applications, or natural disasters

- Malicious or not, a breach of security severely damages the **system's credibility**



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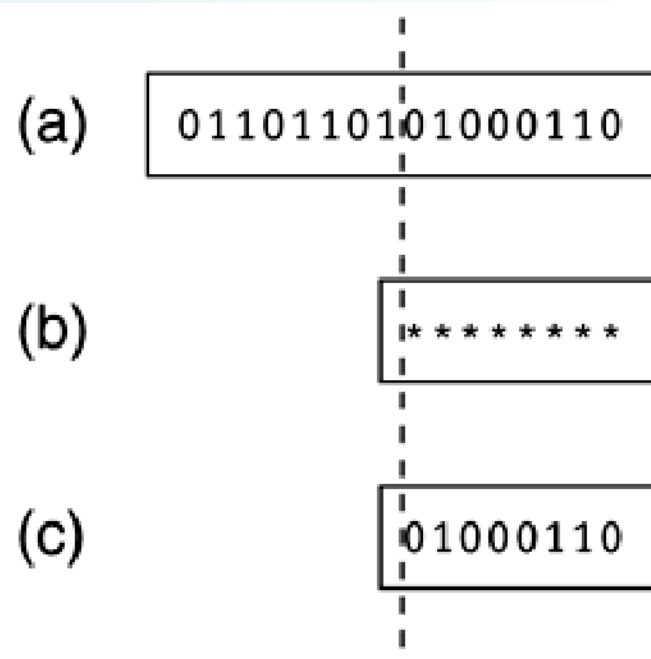
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UNINTENTIONAL INTRUSIONS

- Any breach of security or modification of data that was not the result of a planned intrusion
- **Examples:**
 - Accidental incomplete modification of data
 - When non-synchronized processes access data records and modify some but not all of a record's fields
 - Errors due to incorrect storage of data values
 - Example:** When the field isn't large enough to hold the numeric value stored there



UNINTENTIONAL INTRUSIONS



- (a) Original data value in a field large enough to hold it. If the field is too small,
- (b) FORTRAN replaces the data with asterisks,
- (c) COBOL truncates the higher order digits and stores only the digits that remain

INTENTIONAL ATTACKS

- **Types of Intentional attacks:**

- Intentional unauthorized access

Examples: denial of service attacks, browsing, wire tapping, repeated trials, trap doors, and trash collection

- Viruses and worms
 - Trojan Horses
 - Bombs
 - Blended threats



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INTENTIONAL UNAUTHORIZED ACCESS

- **Denial of service (DoS) attack** is one in which a malicious hacker takes over computers via the Internet and causes them to flood a target site with demands for data and other small tasks causing a computer to perform repeated unproductive task.
- **Browsing** is when unauthorized users gain access to search through secondary storage directories or files for information they should not have the privilege to read.
- **Wire Tapping** Unauthorized users monitor or modify a user's transmission



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INTENTIONAL UNAUTHORIZED ACCESS

- **Repeated Trials** refer to entering systems by guessing authentic passwords
- **Trap doors** refer to an unspecified and undocumented entry point to the system
 - Installed by a system diagnostician or programmer for future use
 - Leaves the system vulnerable to future intrusion
- **Trash collection** refers to the use of discarded materials such as disks, CDs, printouts, etc., to enter the system illegally.



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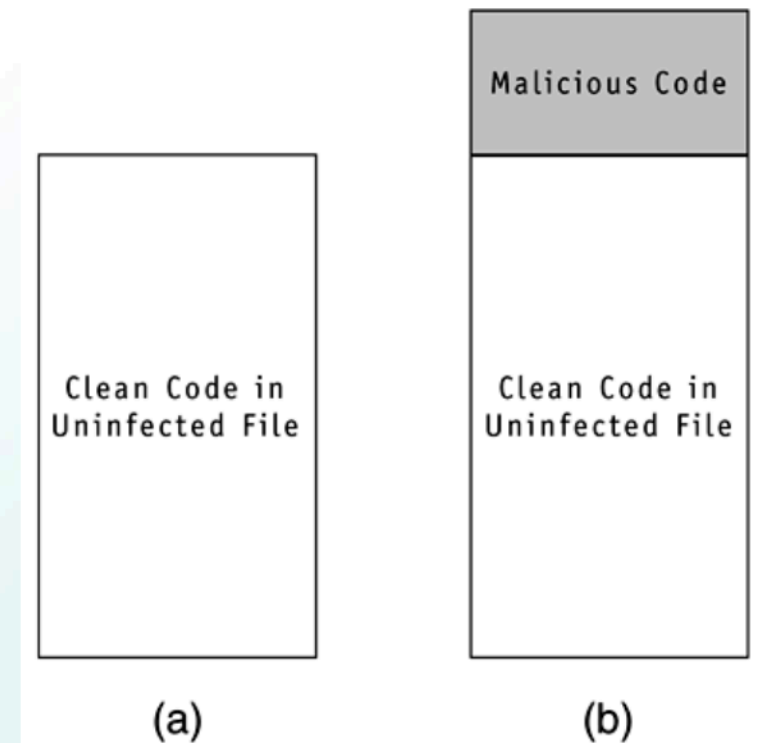
INTENTIONAL UNAUTHORIZED ACCESS

No. of Alphabetic Characters	Possible Combinations	Human Attempt avg. time to discovery at 1 try/second	Computer Attempt avg. time to discovery at 1 million tries/second
1	26	13 seconds	.000013 seconds
2	$26^2 = 676$	6 minutes	.000338 seconds
3	$26^3 = 17576$	2.5 hours	.008788 seconds
8	$26^8 = 208,827,064,576$	6640 years	58 hours
10	$26^{10} = (1.4 \times 10)^{14}$	4.5 million years	4.5 years

Average time required to guess passwords up to ten alphabetic characters (A-Z) using brute force

VIRUSES

- Small programs written to alter the way a computer operates, without permission of the user
- Must meet two criteria: It must be **self-executing** and **self-replicating**
- Usually written to attack a certain OS
- Spread via a wide variety of applications
- **Macro virus** works by attaching itself to a template (such as NORMAL.DOT), which in turn is attached to word processing documents



A file infector virus attacks a clean file (a) by attaching a small program to it (b)

TYPES OF VIRUS

Type of Virus	Description
File infector virus	Infects files on the computer, normally executable files such as .exe and .com files commonly found on Microsoft operating systems. These viruses commonly become resident in memory and then infect any clean executable program that runs on that computer.
Boot sector virus	Infects the boot record, the system area of a floppy disk or hard drive. These viruses activate whenever the user boots up (powers on) the computer. Most boot sector viruses were written for MS-DOS, but other operating systems are potential targets.
Master boot record virus	Infects the boot record of a disk, saving a legitimate copy of the master boot record in a different location on the volume.



TYPES OF VIRUS

Type of Virus	Description
Multipartite virus	Infests both the boot record and program files, making them especially difficult to repair. Successful removal requires that all instances of the virus be removed at once, on the boot records as well as all instances of files infected with the virus. Should any instance of the infection remain, the virus will infect the system again.
Macro virus	Infests data files (such as word processing documents, spreadsheets, etc.), though newer versions now infect other program files as well. Computer users are advised to disable the automatic execution of macros on files they don't completely trust.



BOMBS AND BLENDED THREATS

- **Logic bomb** is a destructive program with a fuse
 - a certain triggering event (such as a keystroke or connection with the Internet)
 - Spreads unnoticed throughout a network
- **Time bomb** is a destructive program triggered by a specific time, such as a day of the year
- **Blended Threat** combines into one program the characteristics of other attacks

Examples: virus, worm, Trojan Horse, spyware, and other malicious code into a single program



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SNIFFERS AND SPOOFING

- **Sniffers** are programs that reside on computers attached to the network. Peruse data packets as they pass by, examine each one for specific information
- **Spoofing** is the act of disguising a communication from an unknown source as being from a known, trusted source.
 - Spoofing can apply to emails, phone calls, and websites
 - Used when unauthorized users want to disguise themselves as friendly sites (**hoax sites**)



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SOCIAL ENGINEERING

A technique whereby system intruders gain access to information about a legitimate user to learn active passwords by

- Looking in and around the user's desk for a written reminder
- Trying the user logon ID as the password
- Searching logon scripts
- Telephoning friends and co-workers to learn the names of user's family members, pets, vacation destinations, favorite hobbies, car model, etc.



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SOCIAL ENGINEERING

- **Phishing** is the act of fraudulently using email to try to get the recipient to reveal personal data. In a phishing scam, con artists send legitimate-looking emails urging the recipient to take action to avoid a negative consequence or to receive a reward.
- **Spear-phishing** is a variation of phishing in which the phisher sends fake emails to a certain organization's employees. It is known as spear-phishing because the attack is much more precise and narrow, like the tip of a spear.
- **Smishing** is a type of phishing that involves the use of Short Message Service (SMS) texting. In a smishing scam, people receive a legitimate-looking text message on their phone telling them to call a specific phone number or to log on to a Web site.



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SOCIAL ENGINEERING

- **Vishing** is similar to smishing except that the victims receive a voice mail telling them to call a phone number or access a Web site.
- **Rootkits** is a set of programs that enables its user to gain administrator-level access to a computer without the end user's consent or knowledge. Once installed, the attacker can gain full control of the system.
- **Ransomware** is a malware that disables a computer or smart-phone until the victim pays a fee, or ransom.



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SYSTEM PROTECTION

- No single guaranteed method of protection
- **System vulnerabilities** include:
 - File downloads, e-mail exchange
 - Vulnerable firewalls
 - Improperly configured Internet connections, etc.
- Need for continuous attention to security issues
- System protection is multifaceted **protection methods** include:
 - Use of antivirus software, firewalls, restrictive access and encryption



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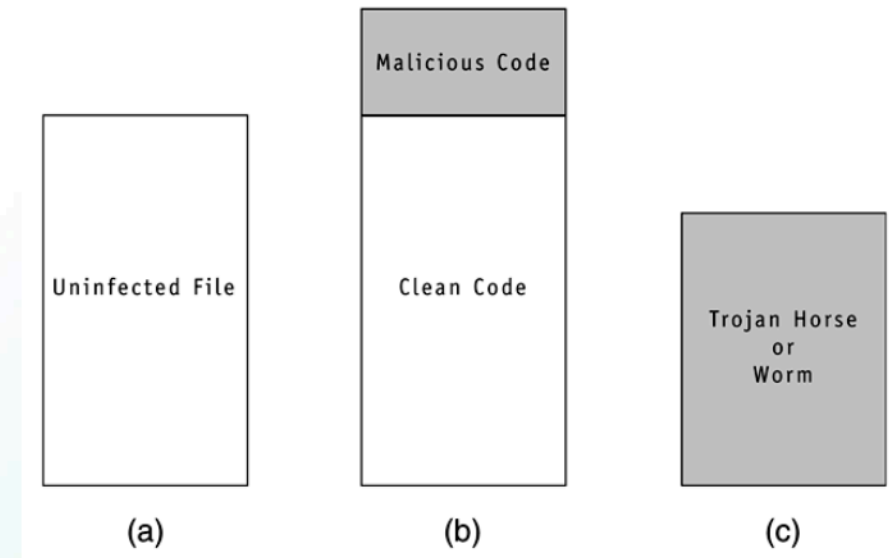


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ANTIVIRUS SOFTWARE

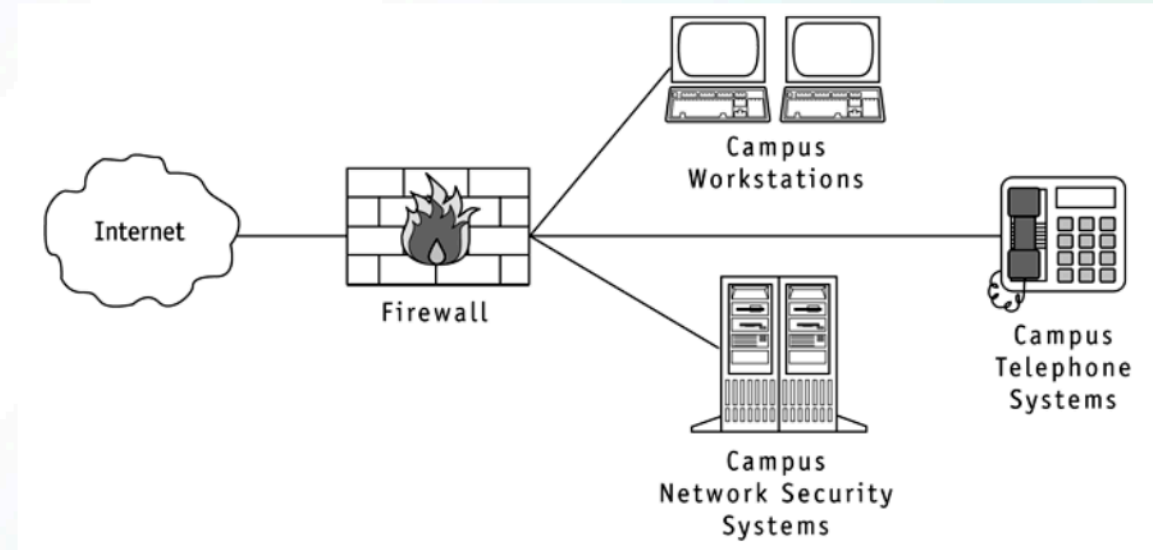
- Software to combat viruses that can be preventive, diagnostic, or both
 - Preventive programs may calculate a checksum for each production program
 - Diagnostic software compares file sizes, looks for replicating instructions or unusual file activity
- Can sometimes remove the infection and leave the remainder intact
- Unable to repair worms, Trojan horses, or blended threats as they are malicious code in entirety



- (a) Uninfected file;
- (b) file infected with a virus;
- (c) a Trojan horse or worm consists entirely of malicious code

FIREWALLS

- A set of hardware and/or software designed to protect a system by disguising its IP address from unauthorized users
- Sits between the Internet and network
- Blocks curious inquiries and potentially dangerous intrusions from outside the system



Firewall sitting between campus networks and Internet, filtering requests for access

FIREWALLS

- Mechanisms used by the firewall to perform various tasks include:
 - Packet filtering
 - Proxy servers
- **Typical tasks of the firewall are to:**
 - Log activities that access the internet
 - Maintain access control based on senders' or receivers' IP addresses
 - Maintain access control based on services that are requested
 - Hide internal network from unauthorized users
 - Verify that virus protection is installed and enforced
 - Perform authentication based on the source of a request from the Internet



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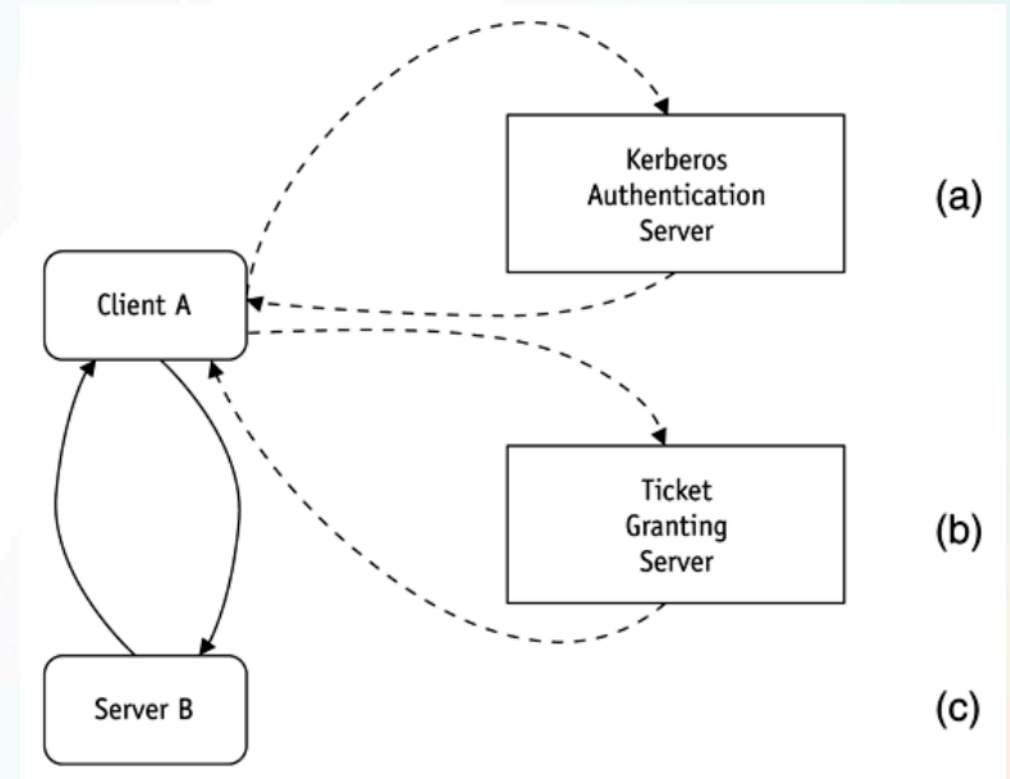


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AUTHENTICATION

- **Authentication** is a verification that an individual trying to access a system is authorized to do so.
- **Kerberos** is a network authentication protocol.
- Designed to provide strong authentication for client/server applications by using secret-key cryptography.



Using Kerberos, when client A attempts to access server B, **user is authenticated (a)**, and **receives a ticket for the session (b)**. Once the ticket is issued, **client and server can communicate at will (c)**. Without the ticket, access is not granted.

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ENCRYPTION

Most extreme protection method for sensitive data where data is put into a secret code

- To communicate with another system, data is encrypted, transmitted, decrypted, and processed
- Sender inserts **public key** with the message
- Message receiver required to have **private key** to decode the message

Disadvantages:

- Increases system's overhead
- System becomes totally dependent on encryption process itself



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PASSWORD MANAGEMENT

Most basic techniques used to protect hardware and software investments include:

- Good passwords
- Careful user training

Password Construction:

- Good password is unusual, memorable, and changed often
- Password files normally stored in encrypted form
- Password length has a direct effect on the ability of password to survive password cracking attempts



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PASSWORD CONSTRUCTION

Reliable techniques for generating a good password:

- Use minimum of eight characters, including numbers and non-alphanumeric characters
- Create a misspelled word or join bits of phrases into a word that's easy to remember
- Follow a certain pattern on the keyboard
- Create acronyms from memorable sentences
- Use upper and lowercase characters if allowed
- Never use a word that's included in any dictionary



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PASSWORD CONSTRUCTION

- **Dictionary attack** is a method of breaking encrypted passwords
 - **Requirements:**
 - A copy of the encrypted password file
 - Algorithm used to encrypt the passwords
 - **Prevention:**
 - Some operating systems “salt” user passwords with extra random bits to make them less vulnerable to dictionary attacks



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PASSWORD ALTERNATIVES

- **Use of a smart card**
 - A credit card-sized calculator that requires both “something you have and something you know”
 - User must type in the number that appears at that moment on the smart card
 - For added protection, user then enters a secret code
 - User is admitted to the system only if both number and code are validated



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PASSWORD ALTERNATIVES

- **Biometrics**
 - The science and technology of identifying individuals based on unique biological characteristics of each person such as human face, fingerprints, hand measurements, iris/retina, and voice prints
 - Positively identifies the person being scanned
 - Critical factor is reducing the margin of error



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